

Ground robot navigation in dense forests with 3D LiDAR using Reinforcement Learning

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Abstract—Navigating autonomous ground robots through dense, unstructured environments like forests remains a significant challenge due to the complexity of the terrain and the computational demands of classical navigation methods. Traditional systems often depend on detailed maps, rigid rule-based frameworks, and sensor fusion strategies that are computationally intensive and lack adaptability across diverse terrains.

This work proposes a reinforcement learning (RL) approach for ground robot navigation in dense forest environments using only a 3D LiDAR sensor and a compass. The RL model, trained using Proximal Policy Optimization (PPO), leverages curriculum learning within MIDGARD, a photorealistic simulation environment specifically designed for outdoor navigation. The training process involves incrementally increasing obstacle density to progressively enhance the agent’s navigation capabilities.

Our experiments demonstrate that the RL-trained agent can efficiently navigate complex forest scenarios, successfully reaching targets despite high obstacle densities. Performance evaluation using Success Rate and Success Path Length (SPL) metrics indicates robust and adaptive navigation behaviors, highlighting the RL model’s potential for real-world deployment on low-power robotic platforms.

Index Terms—unmanned ground vehicle, reinforcement learning, autonomous navigation, outdoor unstructured environments, 3D LiDAR



Fig. 1: Dense Forest Simulation Environment